Ch 12.1, 12.4: Unsupervised Learning & Clustering

Lecture 32 - CMSE 381

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Mon, Nov 25, 2024

Announcements

Last time:

- Convolutional Neural Nets

This lecture:

- Clustering (Just hierarchical clustering)

Announcements:

- No more homework!
- Weds: Project office hours, zoom only, send a message on slack!
- Mon Dec 2: Review - Bring questions!
- Weds Dec 4: Exam
  - Content since 2nd Exam (Ch 7 and on)
  - One page (8.5x11) handwritten cheat sheet
  - Calculator if you want it

Lec #	Date			Reading	HW
21	Mon	10/28	Polynomial & Step Functions	7.1,7.2	
22	Wed	10/30	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
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33	Mon	11/25	TBD: Unsupervised learning/clustering	12.1, 12.4?	
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Section 1

Unsupervised learning

Supervised vs Unsupervised Learning

Supervised

- Have input training data
- Have output labels for that data
- Regression or classification
- Get output $\hat{y}$, compare to actual y

Unupervised

- Don't have training labels
- Goal is more subjective
- Often used in exploratory data analysis
- No universally accepted measure of assessment

Some examples of unsupervised problems

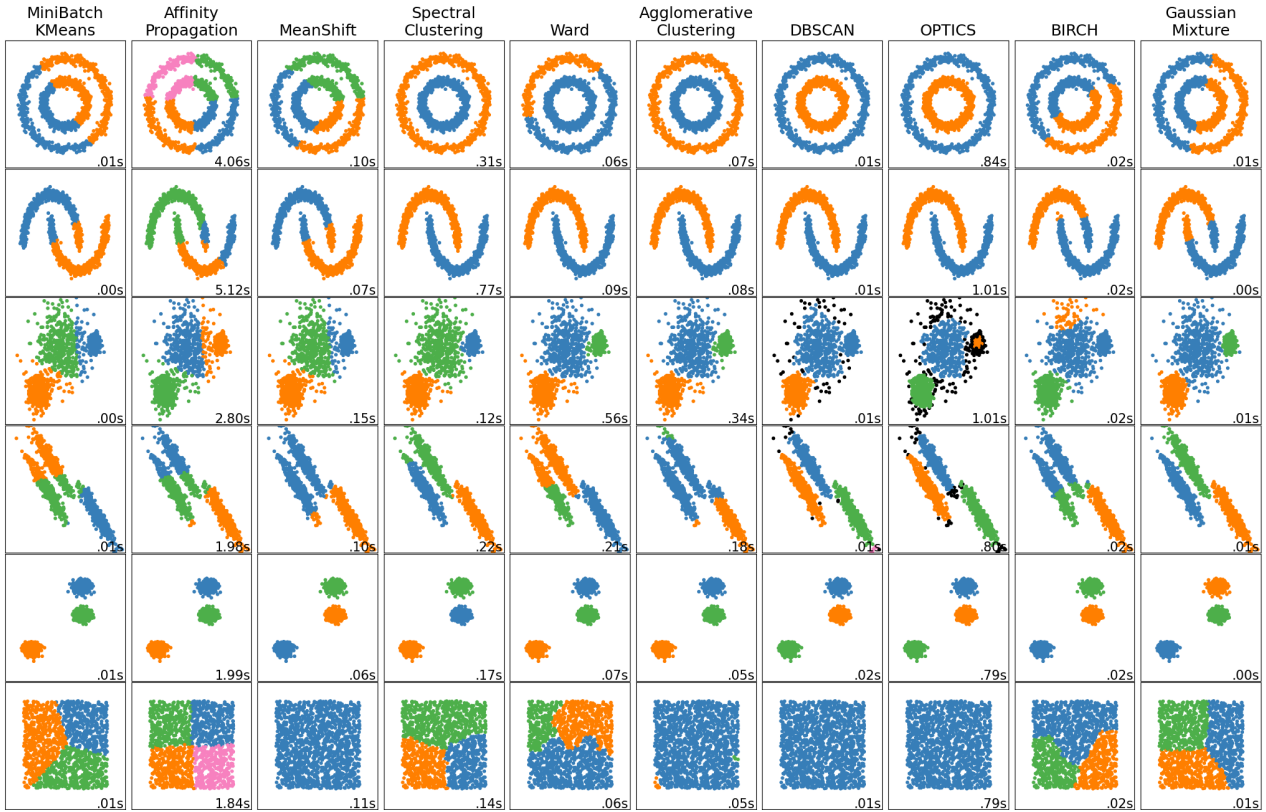
- Assay gene expression levels in 100 patients with breast cancer, looking for subgroups with similar qualities
- Online shopping: find groups of shoppers with similar browsing and purchase histories and show relevant related products.
- Search engine picking results to show

Section 2

Clustering

Big idea

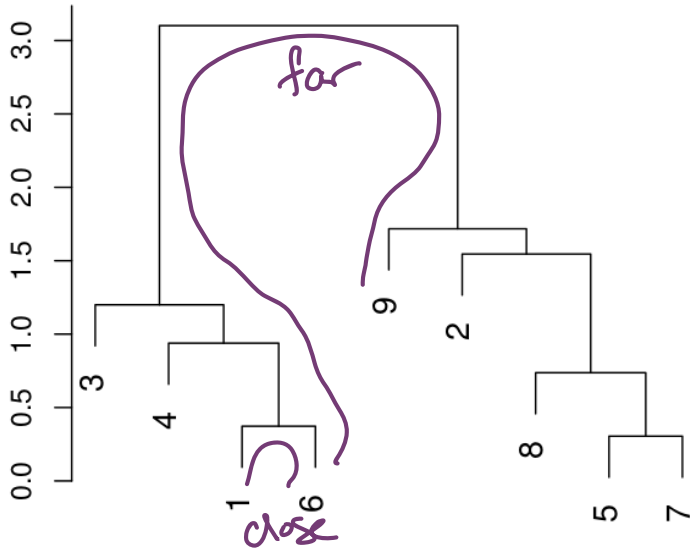
- Find clusters of related data pts in the data set
- Sometimes, ask for # of clusters in advance
- Dependent on what it means for two data pts to be "similar"

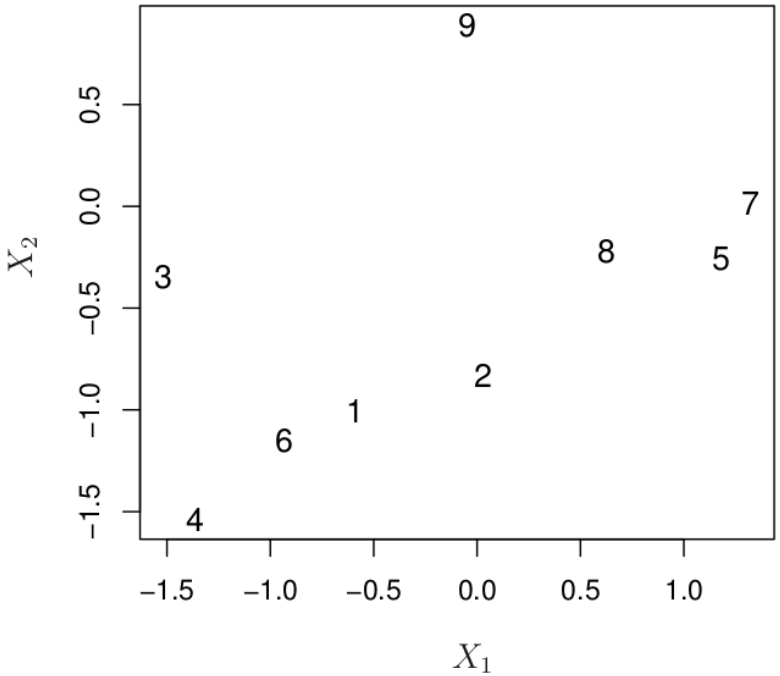


Section 3

Hierarchical Clustering

Dendrogram

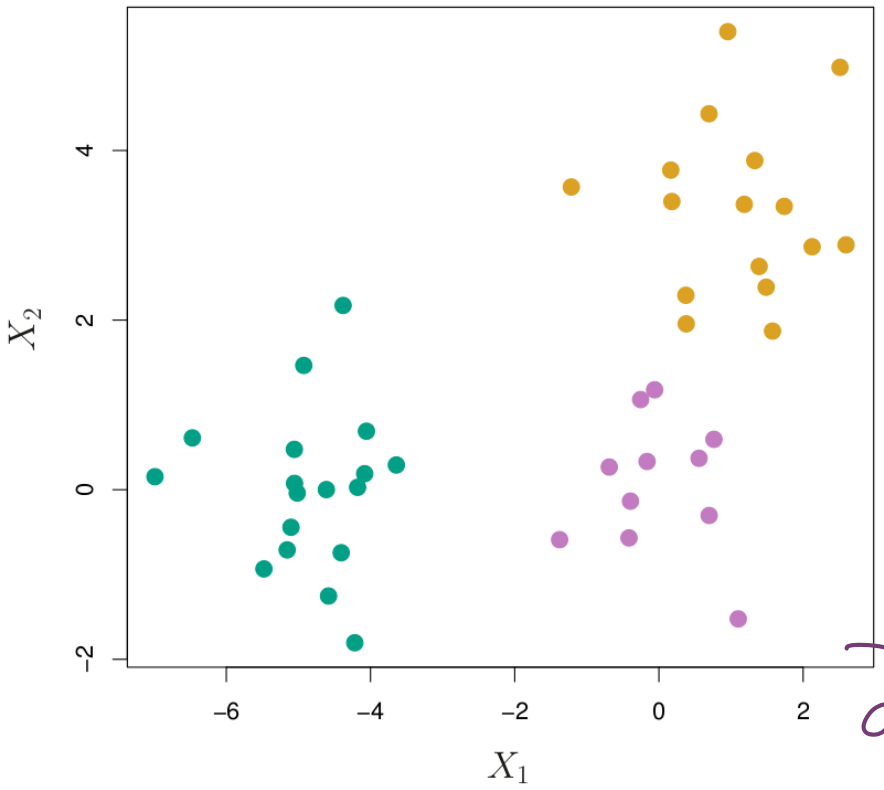




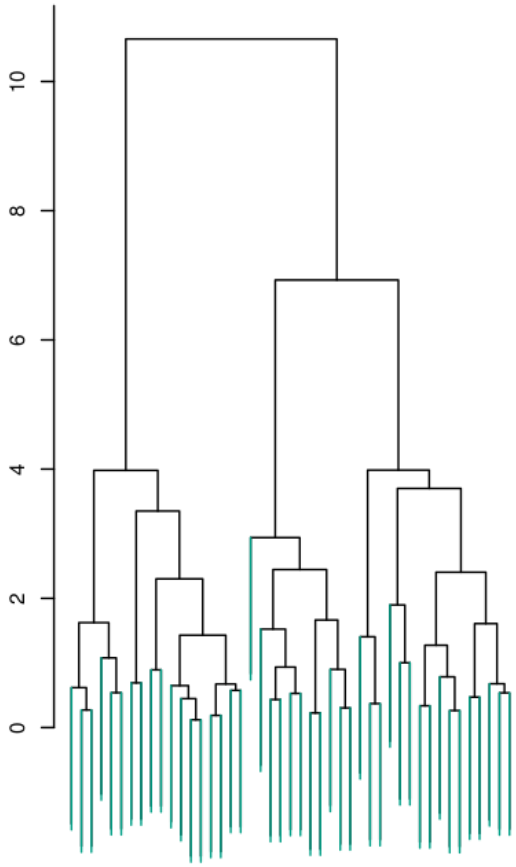
Similar data pts "close" in the tree

Horizontal axis doesn't matter

A bigger example



data pts



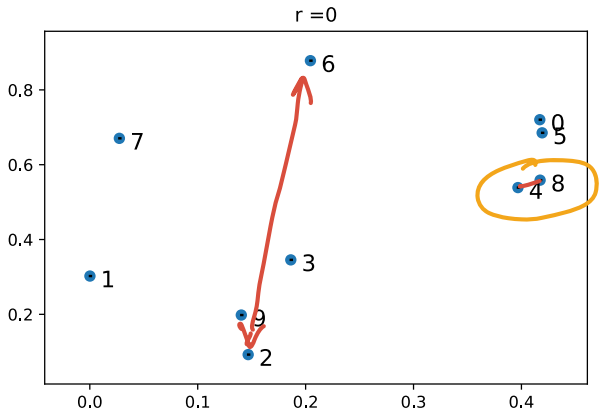
leaves of the tree

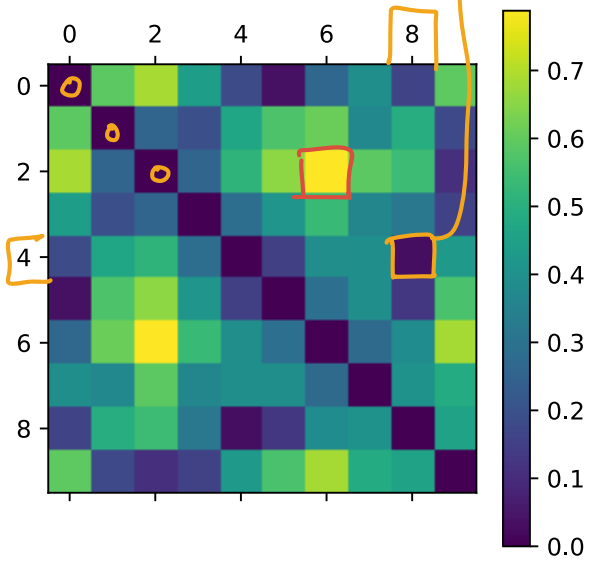
Single linkage

Distance between cluster A and cluster B:
Smallest distance between the points

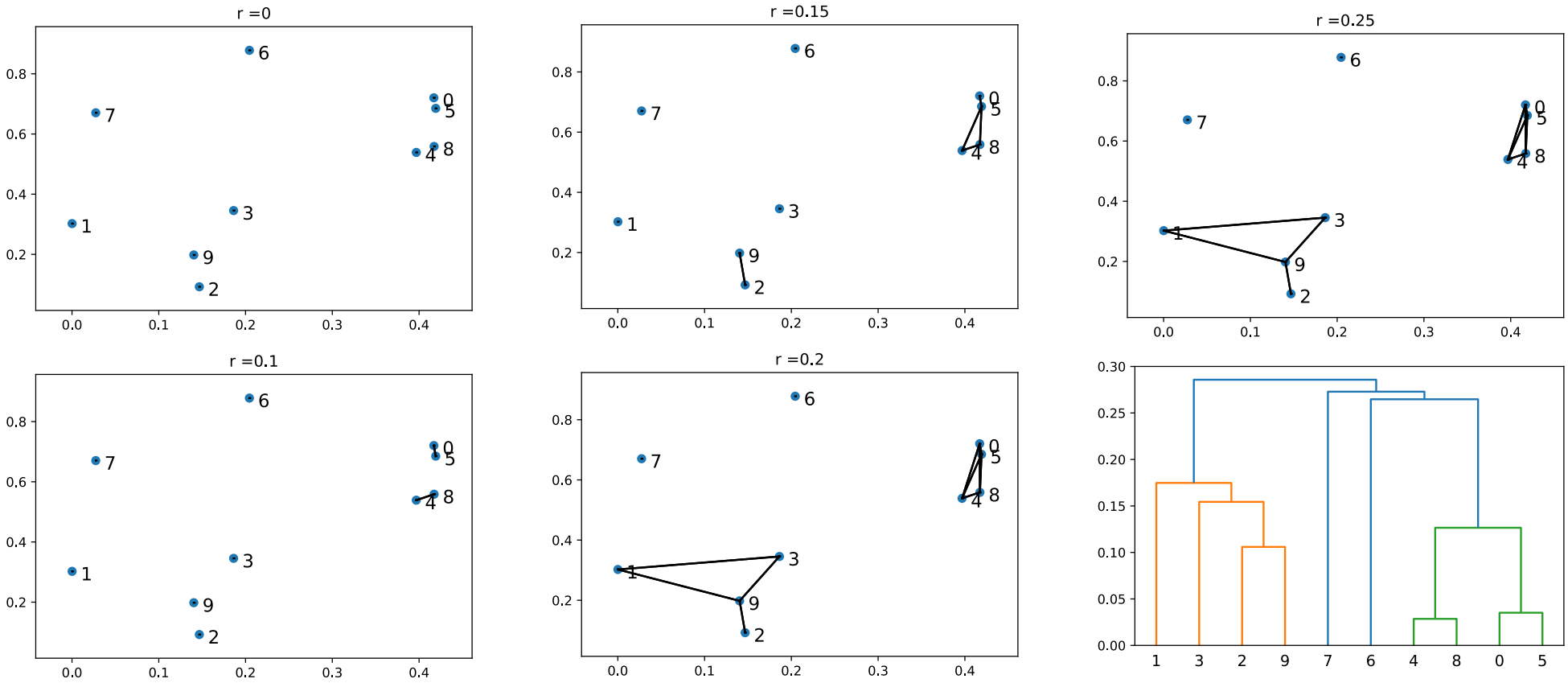
$$L(A, B) = \min_{a \in A, b \in B} \|a - b\|$$

distance from pt 4 to pt 8
$\|a-b\|$

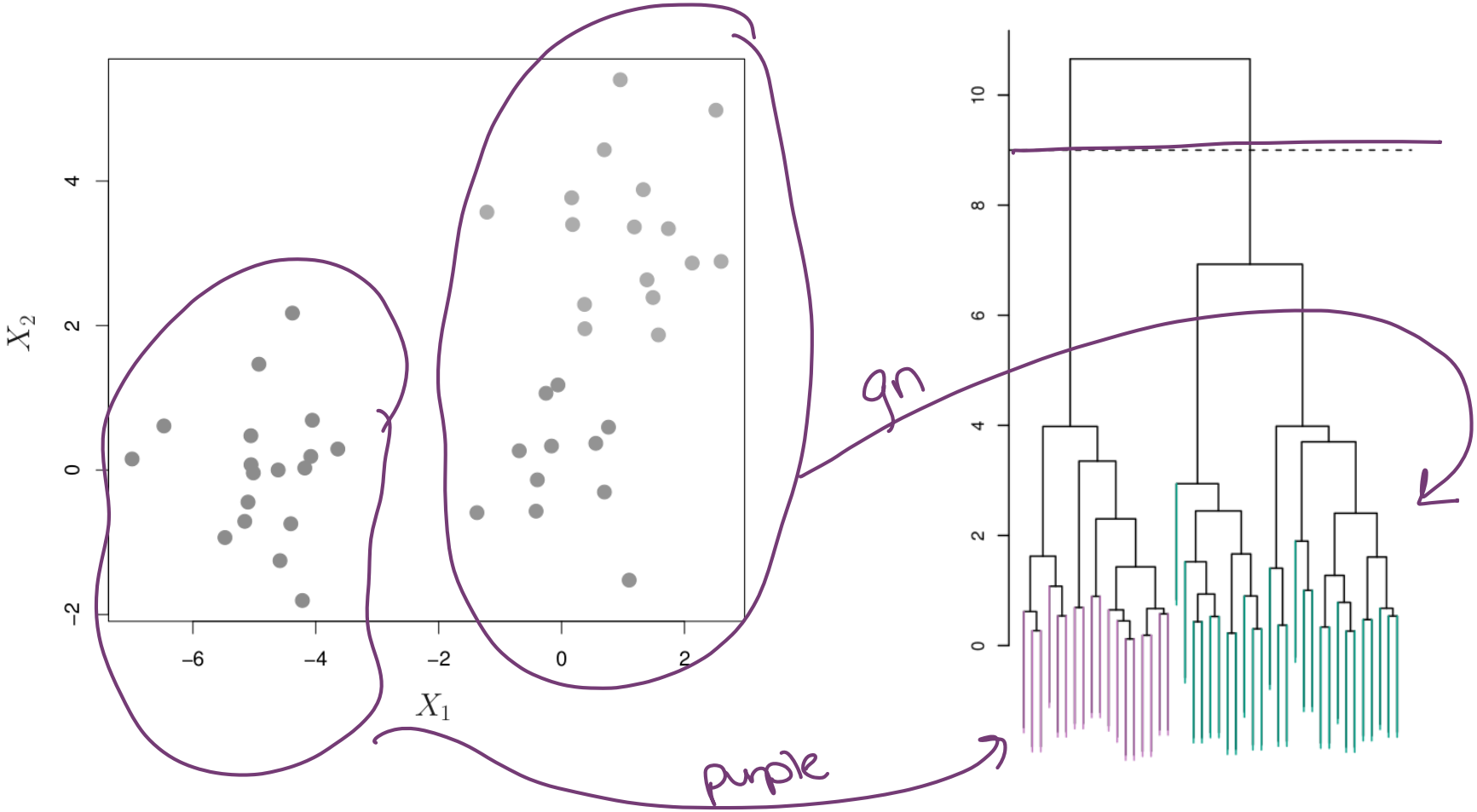




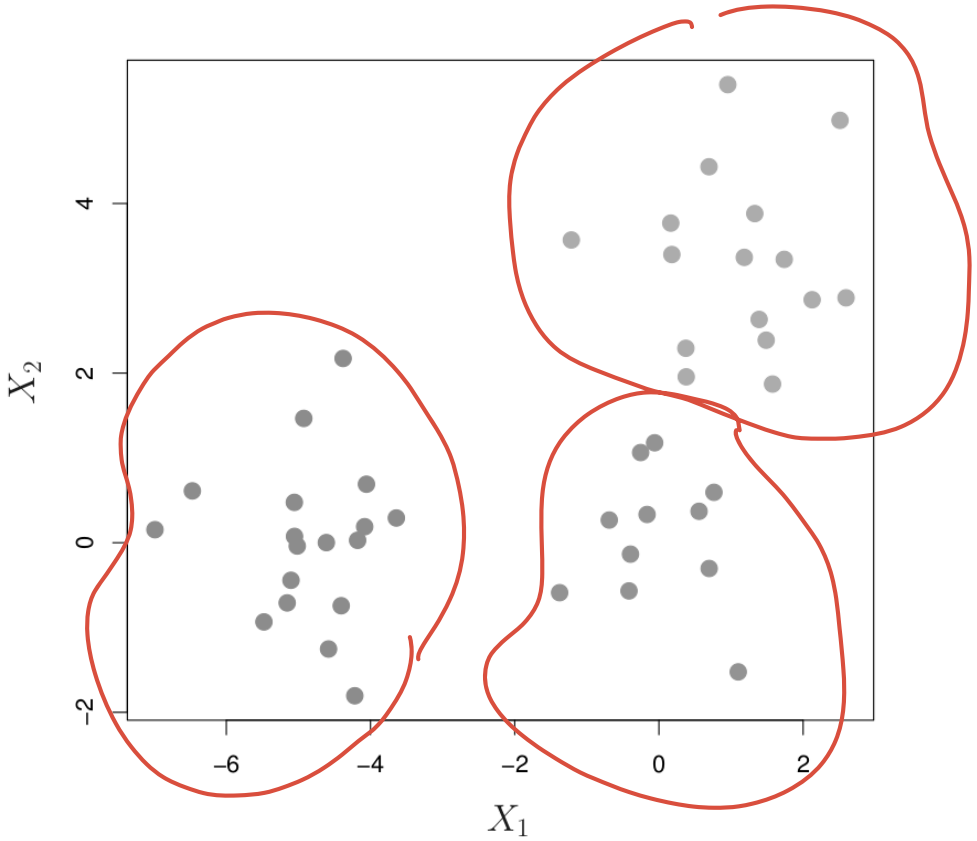
Building the dendrogram

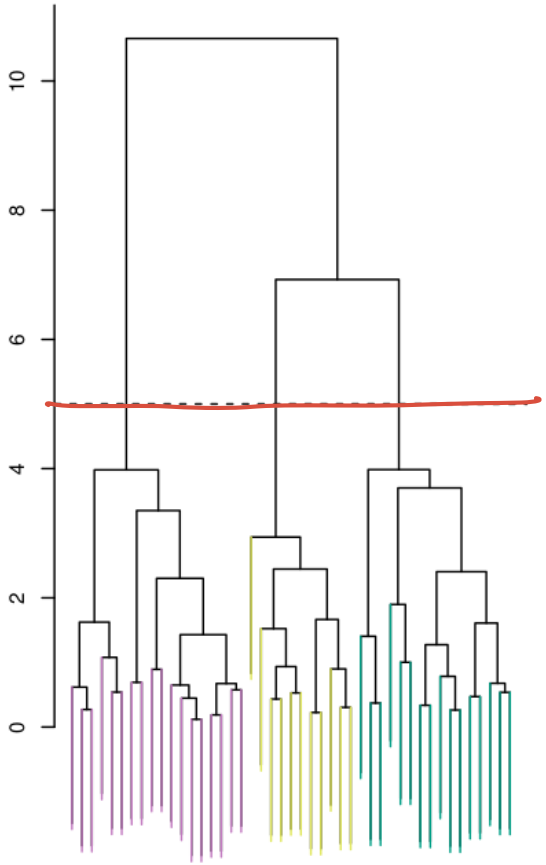


How to get clusters

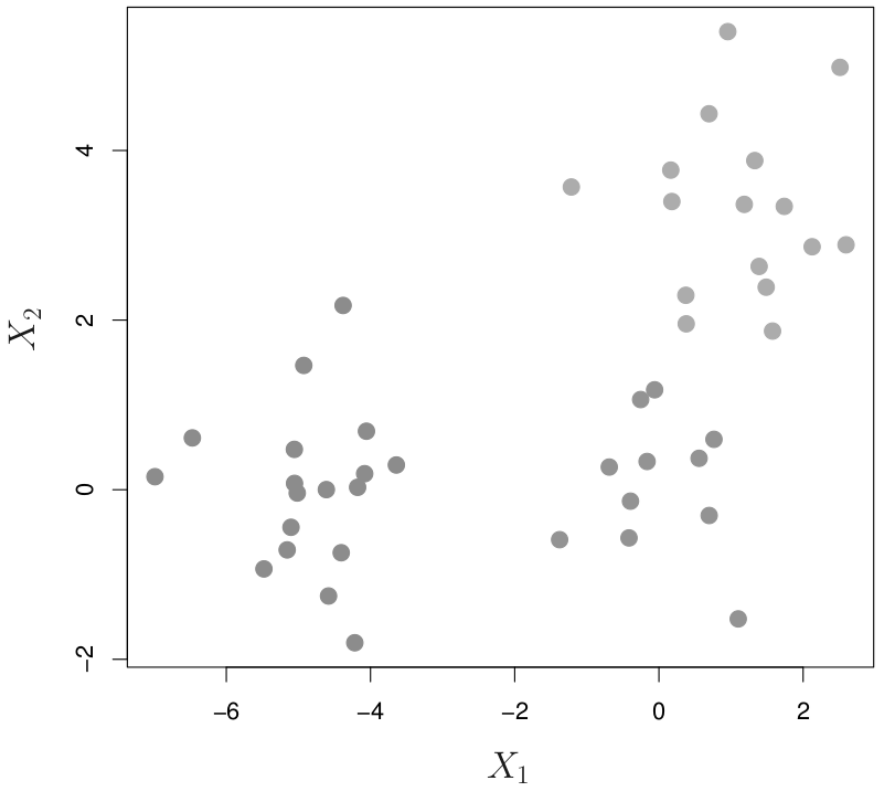


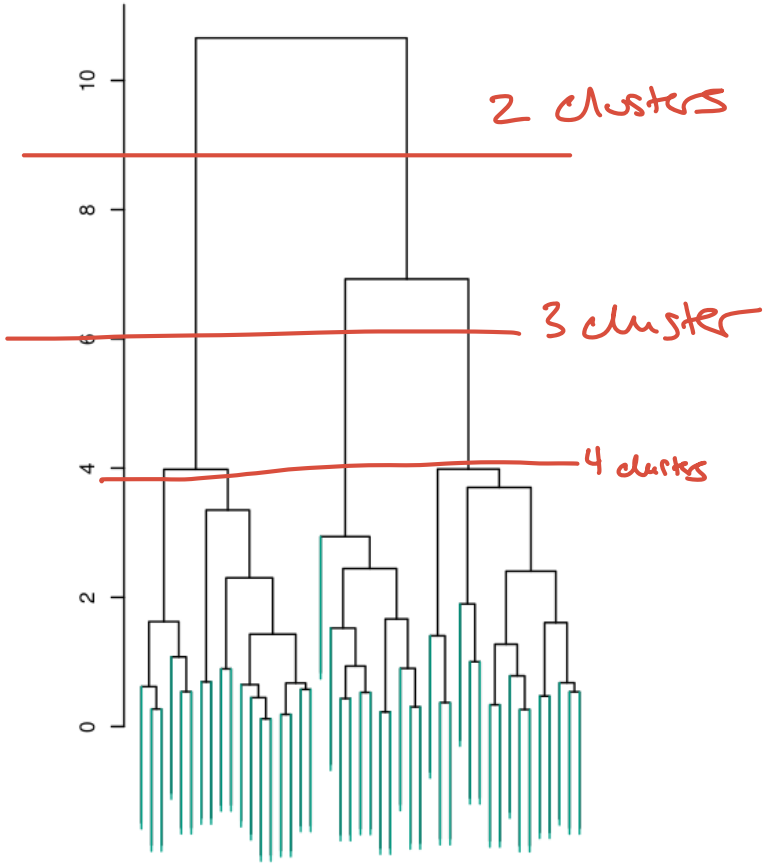
How to get different clusters





Can get any number of clusters





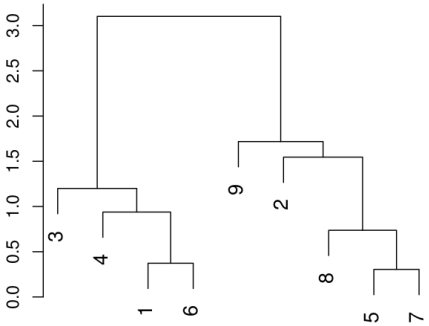
Linkage

Linkage	*Description*
Complete	Maximal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the *largest* of these dissimilarities.
Single	Minimal intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the *smallest* of these dissimilarities. Single linkage can result in extended, trailing clusters in which single observations are fused one-at-a-time.
Average	Mean intercluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the *average* of these dissimilarities.
Centroid	Dissimilarity between the centroid for cluster A (a mean vector of length p) and the centroid for cluster B. Centroid linkage can result in undesirable *inversions*.

prev. example

- lots of options
  get diff outputs
- average and complete tend to have more balanced trees
- Centroid (genomics)
  can create inversion
  two clusters merge @ height below individual pieces

Example with complete linkage



Distance between cluster A and cluster B:

Largest distance between the points

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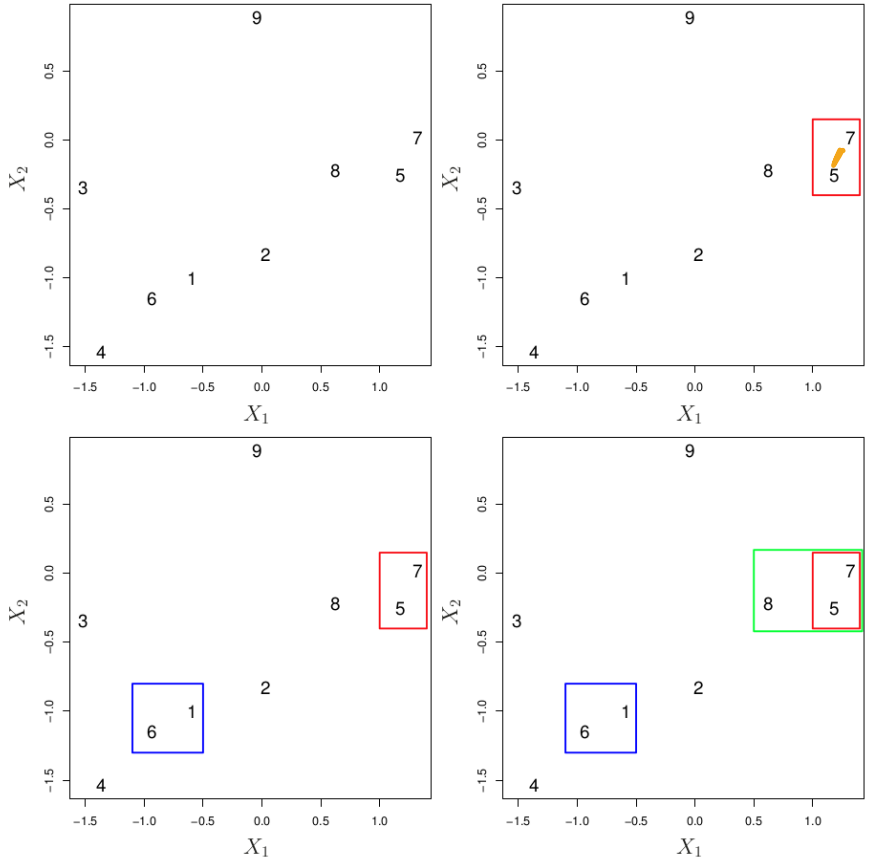
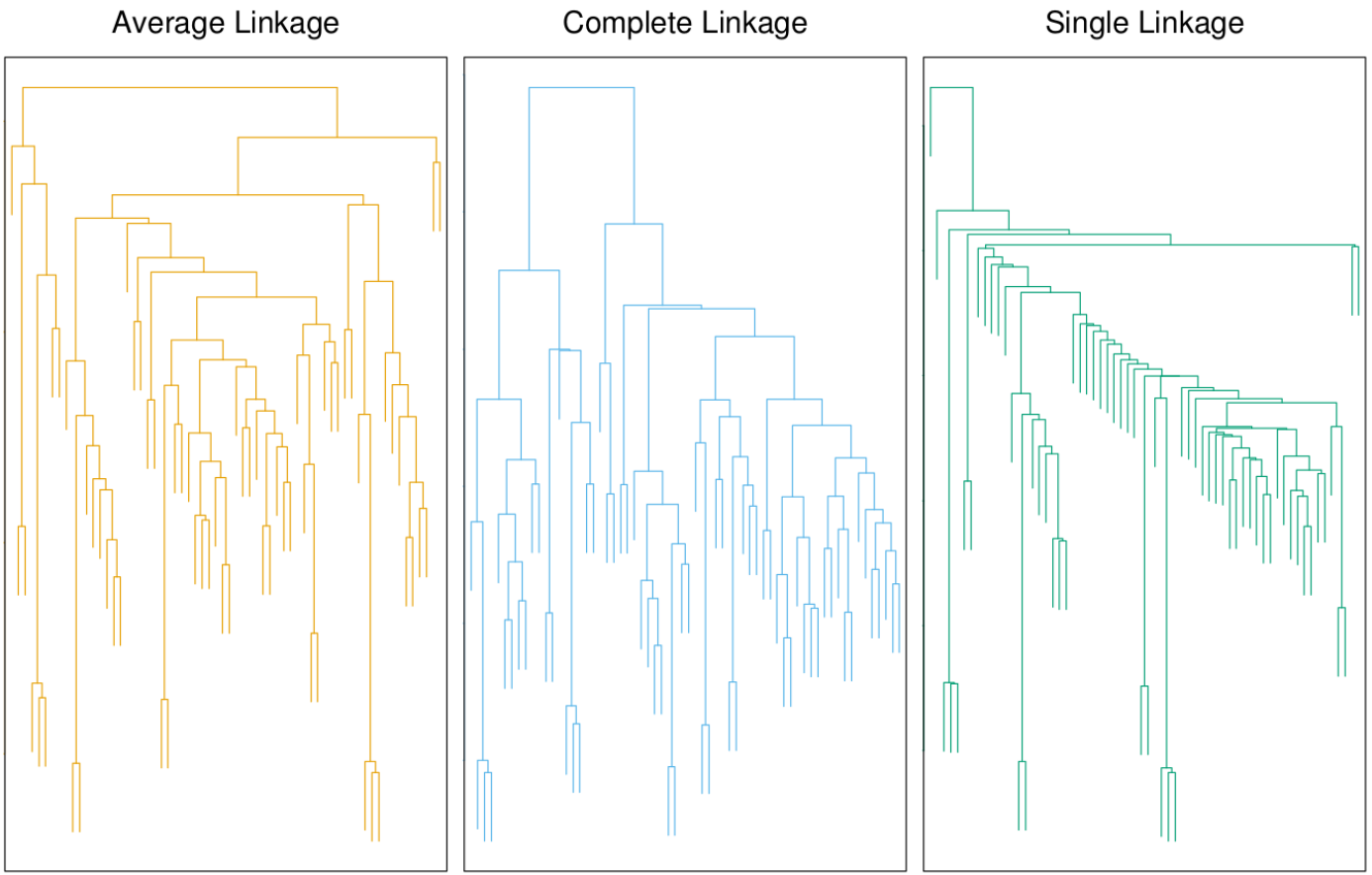


Examples of different linkage



Dependence on dissimilarity measure

Choice is context dependent

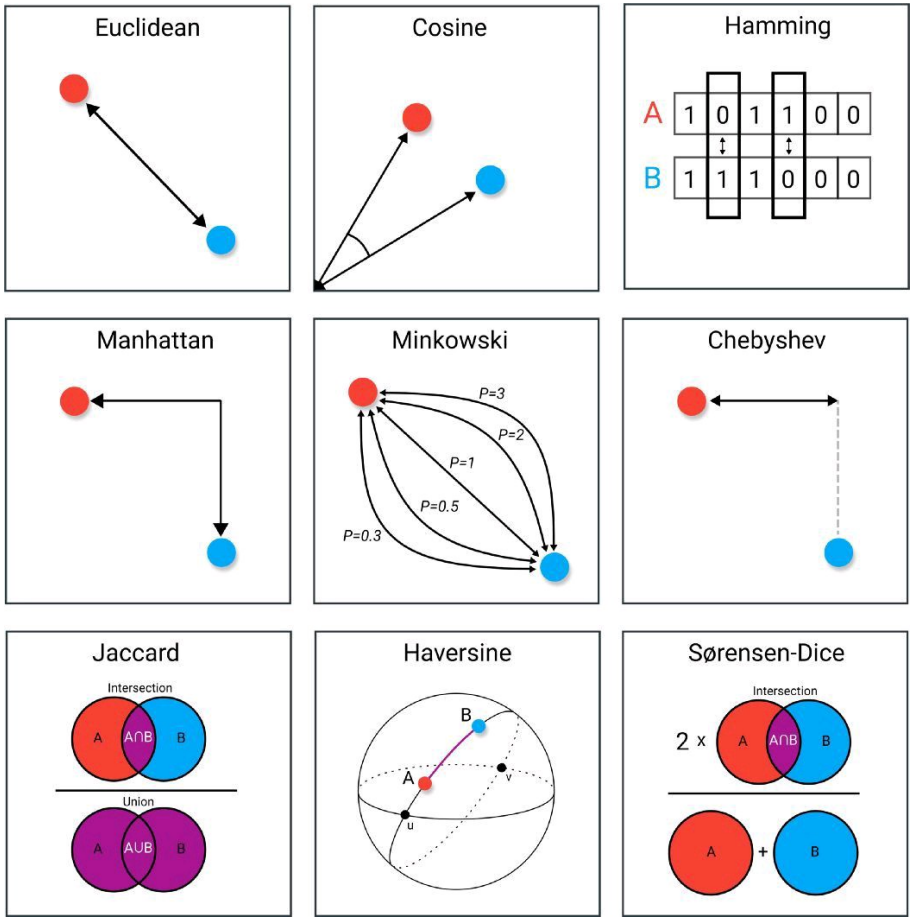


Photo Credit Link

Coding

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